Priyal Jain

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Education

University of Washington (GPA: 3.93/4) — Seattle, WA
Master of Science in Data Science — *Sep 2025 – Mar 2027*
Relevant Coursework: Statistical ML, Probability, Scalable Data Systems, XAI, Software Design, Data Visualization

Vellore Institute of Technology (GPA: 3.72/4) — Amaravati, India
Bachelor's Computer Science & Engineering (Specialization in Data Analytics) — *Jul 2019 – Jul 2023*

Experience

AI Engineer Intern — Jun 2026 – Present
University of Washington IT – Purple AI — *Seattle, USA*

- Designed and evaluated **A/B experiments** on agent prompt configurations using **prompt engineering**, **MCP integrations**, and **LLM fine-tuning**, lifting response accuracy by **15%** across a campus-wide AI platform serving **65,000+ users**
- Established **LLM-as-judge** evaluation pipeline for **production AI agents**, benchmarking 5 LLMs across 4 quality dimensions and **reducing manual QA review time by 65%** through standardized quality gate for all agent releases
- Led development and management of a **safety-critical Distress Response skill**, running **50+ scenario adversarial evaluations** with pass/partial/fail scoring to achieve an **83% pass rate** using **content filters** and **web grounding**

Machine Learning Engineer — Jan 2023 – July 2025
Bosch Global Software Technologies — *Bengaluru, India*

- Architected a production **RAG** system using **LangChain**-orchestrated pipelines, **FAISS/Chroma** vector stores, and **hybrid retrieval** over a multi-decade corpus of automotive safety standards, delivering a **70% reduction** in manual quality assessment time and **$145K USD** in projected annualized operational savings
- Pioneered a modular **LLM evaluation framework**, benchmarking **Mistral, LLaMA, GPT-4, and Claude-3** using **RAGAS, DeepEval, BERTScore**, and lexical metrics (**ROUGE, BLEU, METEOR**) to standardize model selection across **5+ enterprise use cases**, replacing ad-hoc choices with evidence-based decisions
- Spearheaded a **unified AI platform**, fine-tuning open-source LLMs with **QLoRA, LoRA, and RAG** architectures to cut new use-case onboarding time by **25%** and expand to **3+ production systems**, earning Bosch's internal **Growth Driver Award (2024)** for securing a new business venture for the AI/ML department
- Orchestrated the **MLOps** infrastructure underpinning the enterprise AI platform using **Docker**, **GitHub Actions CI/CD**, and **MLflow** experiment tracking, establishing **unit-testing** standards across LLM service deployments

Projects

UniChunk: Multimodal Agentic RAG System | *Python, LangChain, FAISS, LlamaParse, FastAPI*

- Engineered a **GenAI QA system** for **100+ page PDFs** with **multimodal OCR** (text/tables/images/diagrams) and two-tier retrieval using **FAISS/LangChain RAG**, achieving **BERTScore: 0.90** and **Context Similarity: 0.78** through **agentic orchestration**

QuantCopilot: AI Investment Analysis Agent | *Python, Databricks, PySpark, Llama-3.1 405B*

- Developed an **LLM-powered financial agent** using **Llama-3** and **Databricks Unity Catalog** to deliver explainable Buy/Hold/Sell recommendations with **natural-language SQL screening** on **Fortune 500 stocks** and real-time market data, enforcing strict **non-hallucination constraints**

LLM Evaluation Harness | *Python, MLflow, Ragas, Ollama, LangChain, FastAPI, SQLite, GitHub Actions*

- Automated **regression testing** across Llama 3 and Mistral prompt variants by building a **Ragas-scored** FastAPI evaluation service with **MLflow tracking and a GitHub Actions** gate that blocks underperforming changes

Technical Skills

Programming Languages: Python, SQL, Java, C/C++, R
Machine Learning & AI: Large Language Models (LLMs) (Mistral, Llama, GPT, Claude), Retrieval Augmented Generation (RAG) Systems, LangChain, LangGraph, Agentic AI, LLM Fine-tuning (LoRA/QLoRA), Prompt Engineering, Scikit-learn, TensorFlow, PyTorch, HuggingFace, Model Context Protocol (MCP), LLM Evaluation, Deep Learning
Data Engineering & MLOps: Azure, GCP, MLFlow, Databricks, PySpark, ETL/ELT, Docker, Git, CI/CD, GitHub Actions, FastAPI, Kubernetes, Vector Databases (FAISS, Chroma, Pinecone, Weaviate), Claude Code, GitHub Copilot
Data Science & Tools: PostgreSQL, NoSQL, MongoDB, Bash, Tableau, PowerBI, Streamlit, Pandas, NumPy, Matplotlib, Seaborn, A/B Testing, Predictive Modeling, Hypothesis Testing, Stakeholder Communication, Responsible AI